

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1

Year : 2012

Subject: 394173 Mathematics 3

Section : 15-16

Date: 26 September 2012

Time : 13.00 – 16.00

Name _____ ID _____ Field of Study _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Dictionaries and calculators are permitted.
4. No any electronic communication devices are allowed in the exam room.
5. Students should look at their own papers during the examination.
6. The examination has 10 pages (including this page), has 10 problem and a total score of 105 points.
7. Write all your answers on this exam sheet.

- 1 . A data set $A = \{x_1, x_2, x_3, x_4, x_5\}$ has mean, median, mode and range of 17, 16, 15 and 5 respectively. Find the variance of the data set. (10 points)

2. A set of 60 observations has a mean of 22, a variance of 100. If 10 of the 60 observations have the value of 32 each, find the variance of the remaining 50 observations. (10 points)

3. In order to calibrate 2 balances A and B, a 10 kg object has been weighed 6 times. Each time the weight has been recorded as shown in table A. (10 points)

	Weight recorded (g)	
	Balance A	Balance B
1	9	10
2	9.9	9.5
3	9.2	9.3
4	9.1	8.5
5	10	7.2
6	7.8	9.0

Find the coefficient of variance of two groups. Which group is the best?

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4. From the following data

class	Cumulative frequency
65-79	5
50-64	23
35-49	37
20-34	47
5-19	50

Calculate the values of Mean, Median, and Mode? (10 points)

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5. The following data is given by

class	frequency
10-19	3
20-29	5
30-39	7
40-49	12
50-59	13
60-69	4
70-79	2

Find the value of quartile deviation and the coefficient of variation of the data. (10 points)

6. If mathematical scores are normally distributed with a mean of 80 and a standard deviation of 15.

6.1 What is the probability that a person chosen at random has a score greater than 95?

(3 points)

6.2 What is the probability that a person chosen at random has a score less than 60?

(3 points)

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6.3 In order to join a mathematical club one has to have an score in the top 5% of all the population. What the score would be necessary to join this club? (4 points)

7. Heights of adult women in the United States are normally distributed with a population mean 63.5 inches and a standard deviation of 2.5 inches. A medical researcher is planning to select a large random sample of adult women to participate in a future study. What is the standard value, or z-value, for an adult woman who has a height of 68.5 inches? (5 points)

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8. An automobile manufacturer introduces a new model that averages 27 miles per gallon in the city. A person who plans to purchase one of these new cars wrote the manufacturer for the details of the tests, and found out that the standard deviation is 3 miles per gallon. Assume that in-city mileage is approximately normally distributed.
- 8.1 What is the probability that the person will purchase a car that averages less than 20 miles per gallon for in-city driving? (5 points)

8.2 What is the probability that the person will purchase a car that averages between 25 and 29 miles per gallon for in-city driving? (5 points)

9. In an air pollution study, the following of suspended benzene soluble organic matter for each year were obtained by the table, plot the scattergram. Find the straight line of best fit, and sketch the graph of the line. (10 points)

x	y
2001	1.2
2002	1.8
2003	2.0
2004	2.1
2005	2.2
2006	2.4
2007	3.3

10. The following table shows the quantities and prices of TV and radio products in 2010 and 2012

Lists of product	Price(unit : 1000 Baht)		Quantity (unit : 1000 Baht)	
	2010	2012	2010	2012
TV	5	8	100	105
Radio	3	6	56	56

Calculate price indices by using Laspeyres index method and Paasche index method which the base period is in 2010. (10 points)

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